

STAGE 5 INDUSTRIAL TECHNOLOGY ELECTIVE - METAL INDUSTRY

Folding Camping Shovel | 25-period teaching program

Website-complete teacher review copy | Industrial Technology 7-10 Syllabus (2025)

Course	Stage 5 Industrial Technology elective - Metal industry	Delivery	2027 10 weeks
Timetable	25 x 60-minute periods	Pattern	5 periods per fortnight
Lesson balance	5 theory 5 project/folio 15 practical	Syllabus	2025 syllabus - approved early use
Status	Local teacher review only	Project	Produce the approved folding camping shovel project

Website coverage: 15 named theory sections, 15 quiz/check sets (150 questions) and 15 scaffolded responses are scheduled. Practical lessons remain the majority.

Industrial Technology outcomes

Outcome	Official wording
INT5-SAF-01	applies risk management and safe work practices in industrial technology contexts
INT5-COM-01	communicates ideas and processes in industrial technology contexts
INT5-ENV-01	assesses the impact of industrial activities and practices on society and the environment
INT5-DES-01	selects and applies design, project management and technical skills in project production
INT5-USE-01	selects and justifies the use of materials, tools and equipment to undertake projects
INT5-EVL-01	evaluates the functional, aesthetic, economic and environmental qualities of products
INT5-INP-01	analyses and applies a range of industrial processes in production contexts
INT5-PRO-01	demonstrates practical skills in the production of projects
INT5-IND-01	investigates and explains a range of industry careers, enterprise opportunities and practices

Teacher to confirm before use

When	Confirm	Owner
Before Period 1	Assessment dates, submission method, class supports and adjustments	Teacher / faculty
Before practical work	Current plan revision, permitted variations, induction, SOPs, tools, PPE, supervision and risk controls	Teacher / faculty
Before finishing	Approved finish, SDS/product instructions, area, drying and storage arrangements	Teacher / faculty

Teaching program | Periods 1-10

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
1	Site/theory	Module 1: website	Read and discuss: Reading the folding-shovel system through its drawings; Reading the folding-shovel system through its drawings knowledge check (10 questions); written response. Save or show the completed website evidence.	Accurately measure and mark out metal	Website evidence checked INT5-USE-01, INT5-PRO-01
2	Project/folio	Module 1: folio	Read and complete: WMS planning for a multi-process product; WMS planning for a multi-process product knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Accurately measure and mark out metal	Folio + website evidence INT5-USE-01, INT5-PRO-01
3	Practical	Module 1: site + practical	First 20 min: Societal and environmental criteria before manufacture; Societal and environmental criteria before manufacture knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Sustainable practices in the metal industry	Website + practical check INT5-ENV-01, INT5-IND-01
4	Practical	Module 1: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Accurately measure and mark out metal	Practical quality check INT5-USE-01, INT5-PRO-01
5	Practical	Module 1: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Accurately measure and mark out metal	Photo + quality note INT5-USE-01, INT5-PRO-01
6	Site/theory	Module 2: website	Read and discuss: Translating the shovel-head drawing into a controlled sequence; Translating the shovel-head drawing into a controlled sequence knowledge check (10 questions); written response. Save or show the completed website evidence.	Accurately measure and mark out metal	Website evidence checked INT5-USE-01, INT5-PRO-01
7	Project/folio	Module 2: folio	Read and complete: Material behaviour during forming and shaping; Material behaviour during forming and shaping knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Metal properties and justified material selection	Folio + website evidence INT5-USE-01, INT5-ENV-01
8	Practical	Module 2: site + practical	First 20 min: Checking head geometry, edges and assembly readiness; Checking head geometry, edges and assembly readiness knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Technical skill for quality projects	Website + practical check INT5-PRO-01, INT5-INP-01
9	Practical	Module 2: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Risk management and risk-reduction procedures	Practical quality check INT5-SAF-01, INT5-USE-01
10	Practical	Module 2: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Risk management and risk-reduction procedures	Photo + quality note INT5-SAF-01, INT5-USE-01

Teaching program | Periods 11-20

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
11	Site/theory	Module 3: website	Read and discuss: Metal-lathe SOP evidence for handle and end-plug work; Metal-lathe SOP evidence for handle and end-plug work knowledge check (10 questions); written response. Save or show the completed website evidence.	Risk management and risk-reduction procedures	Website evidence checked INT5-SAF-01, INT5-USE-01
12	Project/folio	Module 3: folio	Read and complete: Functional fits connect dimensions to movement and assembly; Functional fits connect dimensions to movement and assembly knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Accurately measure and mark out metal	Folio + website evidence INT5-USE-01, INT5-PRO-01
13	Practical	Module 3: site + practical	First 20 min: Threads as controlled geometry and evidence; Threads as controlled geometry and evidence knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Cut internal and external screw threads	Website + practical check INT5-INP-01, INT5-PRO-01
14	Practical	Module 3: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Metal properties and justified material selection	Practical quality check INT5-USE-01, INT5-ENV-01
15	Practical	Module 3: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Metal properties and justified material selection	Photo + quality note INT5-USE-01, INT5-ENV-01
16	Site/theory	Module 4: website	Read and discuss: Pivot parts: centres, alignment and movement path; Pivot parts: centres, alignment and movement path knowledge check (10 questions); written response. Save or show the completed website evidence.	Metal joining and fabrication techniques	Website evidence checked INT5-INP-01, INT5-PRO-01
17	Project/folio	Module 4: folio	Read and complete: Milling as a controlled relationship between tool, work and reference; Milling as a controlled relationship between tool, work and reference knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Use machining and fabrication tools and equipment	Folio + website evidence INT5-SAF-01, INT5-INP-01
18	Practical	Module 4: site + practical	First 20 min: Drilling RPM calculations: formula, units and authorised setting; Drilling RPM calculations: formula, units and authorised setting knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Use machining and fabrication tools and equipment	Website + practical check INT5-SAF-01, INT5-INP-01
19	Practical	Module 4: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Use machining and fabrication tools and equipment	Practical quality check INT5-SAF-01, INT5-INP-01
20	Practical	Module 4: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Use machining and fabrication tools and equipment	Photo + quality note INT5-SAF-01, INT5-INP-01

Teaching program | Periods 21-25

P	Type	Focus	Teaching and learning	Syllabus content	Evidence / outcomes
21	Site/theory	Module 5: website	Read and discuss: Handle and pivot assembly as a functional quality gate; Handle and pivot assembly as a functional quality gate knowledge check (10 questions); written response. Save or show the completed website evidence.	Metal joining and fabrication techniques	Website evidence checked INT5-INP-01, INT5-PRO-01
22	Project/folio	Module 5: folio	Read and complete: Societal and environmental impact report with evidence; Societal and environmental impact report with evidence knowledge check (10 questions); written response. Update the folio with the current decision, process evidence and next step.	Sustainable practices in the metal industry	Folio + website evidence INT5-ENV-01, INT5-IND-01
23	Practical	Module 5: site + practical	First 20 min: PMI, project report and authentic Task 2 submission; PMI, project report and authentic Task 2 submission knowledge check (10 questions); written response. Then complete the teacher-approved project stage; check the plan, setup and one quality point.	Technical skill for quality projects	Website + practical check INT5-PRO-01, INT5-INP-01
24	Practical	Module 5: production	Continue the approved project stage; follow the current plan and SOP; check accuracy, fit or surface quality.	Use machining and fabrication tools and equipment	Practical quality check INT5-SAF-01, INT5-INP-01
25	Practical	Module 5: evidence	Complete the current stage; photograph a checkpoint; record the problem and response if needed; clean down.	Use machining and fabrication tools and equipment	Photo + quality note INT5-SAF-01, INT5-INP-01

Evidence, feedback and practical controls

Check	Look for	Respond
Website	Named theory, checks and responses completed	Revisit feedback and correct missed ideas
Folio	Decisions and process evidence tied to the project	Set one clear next step
Practical	Safe method, accuracy, quality and clean-down	Pause, demonstrate, practise and recheck

Use the current approved plan, local induction, SOPs, PPE, supervision, risk controls and product instructions for every practical lesson.